YANG YANG

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Education

South-Central Minzu University — Wuhan, China
M.S. in Computer Technology — *Sep 2024 – Jun 2027 (expected)*
South-Central Minzu University — Wuhan, China
B.E. in Network Engineering — *Sep 2020 – Jun 2024* | *GPA: 3.17/4.0*

Research Interests

Computer Vision, Medical Image Segmentation, Multimodal Learning

Publications

Yang Yang, Yuxin Hong, Wenjie Yu, Xiao Zhang, Bo Yang, Meng Shi, Yangguang Sun, Jun Wang, and Jianlin Zhu*. HSFPN-Det: an effective model for detecting rice pests and diseases. *The Visual Computer (TVC)*, 2025. [Paper]. (**JCR Q2, IF: 3.4**).

Yang Yang, Jianlin Zhu*, Bo Yang, Xiao Zhang, and Jin Huang. An improved YOLOv8-based rice pest and disease detection method. *Computer Graphics International (CGI)*, 2024. [Paper]. (**CCF C**).

Yang Yang, Xuyan Huang, Xiao Zhang, Pan Lai, Bo Yang, Meng Shi, Zimao Li, Jun Tie, Yang Wen, Dian Zeng, Lu Liu, and Jianlin Zhu*. Background as parts: sub-background modeling for uncertainty-aware semi-supervised medical image segmentation. **(Under Review)**.

Yang Yang, Xuyan Huang, Xiao Zhang, Pan Lai, Bo Yang, Meng Shi, Zimao Li, Yang Wen, Dian Zeng, Lu Liu, Yucong Zhang, and Jianlin Zhu*. Learning symmetric cross-modal representations via feature distillation and contrastive fusion for brain tumor segmentation. **(Under Review)**.

Yang Yang, Xuyan Huang, Xiao Zhang, Jianlin Zhu*, Pan Lai, Bo Yang, Meng Shi, Zimao Li, Jun Tie, and Yang Wen. Hi-ERMNet: hierarchical transformer with edge refinement for retinal vessel segmentation. **(Under Review)**.

Xuyan Huang, **Yang Yang**, Yao Lv, Xiao Zhang, Meng Shi, Pan Lai, Bo Yang, Zimao Li, Jun Tie, Dian Zeng, Lu Liu, Bing Sheng. Ontology-Guided Lesion-Aware Vision-Language Alignment for Retinal Image-Text Retrieval. **(Under Review)**.

Research Experience

Graduate Researcher, South-Central Minzu University — Apr. 2025 – Sept. 2025
Nüshu Character Detection and Generation — *Wuhan, China*

- ***Model Development:*** led the construction of the character detection and generation models.
- ***Dataset:*** collaborated with team members, who manually collected a Nüshu character dataset used for model training and evaluation.

Graduate Researcher, South-Central Minzu University — Sept. 2024 – Dec. 2024
Cardiovascular Disease Prediction from Wearable ECG/PPG Signals — *Wuhan, China*

- ***Data Analysis:*** performed preprocessing and analysis of wearable-device-collected ECG and PPG physiological signals.

- ***Time-Series Modeling:*** built time-series prediction models as part of the foundational, early-stage pipeline for cardiovascular disease risk prediction.

Graduate Researcher, South-Central Minzu University Jul. 2024 – Nov. 2024
Data-Mechanism Integrated Virtual Model for Ship Fire Evolution *Wuhan and Shanghai, China*

- ***Fire Evolution Modeling:*** prepared a project proposal that uses Unity 3D to model selected ship compartments, employs PyroSim to analyze fire evolution, and proposes a ship fire evolution model.
- ***Physics-Informed Modeling:*** applied physics-informed neural networks (PINNs) to model ship fire evolution and support early-warning and risk assessment for compartment fire scenarios.

Selected Honors & Awards

- *First-class Academic Scholarship for Graduate Students* Sept. 2025
- *Outstanding Graduate Student* Sept. 2025
- *Second-class Academic Scholarship for Graduate Students* Sept. 2024
- *National Encouragement Scholarship* Sept. 2023

Skills and Software

Programming: Python, C/C++, Java **Tools:** Git, Linux
Scientific Writing: LaTeX **Frameworks:** PyTorch
Data Analysis & Visualization: Numpy, Scipy, Matplotlib

Languages

Chinese Mandarin (native), English

Services

Reviewer: ACM MM, Expert Systems With Applications, CGI, Journal of Computer-Aided Design & Computer Graphics